

Assignment - 3

PROBLEM STATEMENT

Write a C program to create a student database on array of structures. Apply searching (linear and Binary Search) and sorting technique (insertion sort, selection sort, shell sort).

OBJECTIVE

- 1) To understand use of an array structure for maintaining records
- 2) To implement and analyze searching techniques
- 3) To implement and analyze sorting technique
- 4) To compare insertion sort, selection sort, selection sort, shell sort.

Theory

Searching and sorting are two types of the fundamental operations in data structures and algorithms. They are widely used in computer applications.

IMPLEMENTATION

Platform

64-bit open source linux or its derivatives open source C programming tool, like gcc/eclipse editor.

Pseudo code

Write pseudo code linear search, Binary search, Insertion sort, selection and shell sort.

1) Linear search

Algorithm Linear search (array, n, target):
 for i from 0 to $n-1$:
 If array [i] == target
 Return i
 Return -1

2) Binary search

Algorithm Binary search (array, low, high, target)

$$mid = low + (high - low) / 2$$

If array [mid] == target
 Return mid

Else if array [mid] < target:
 $low = mid + 1$

Return -1

3) Insertion sort

Algorithm Insertion sort (array, n):

for i from 1 to $n-1$:

key = array [i]

$j = i - 1$

while $j > 0$ and array [j] > key

array [$j+1$] = array [j]

$j = j - 1$

array $[i + 1] = \text{key}$

4) Selection sort

Algorithm selection sort (array, n):

for i from 0 to $n - 2$:

min index = 1

for j from $i + 1$ to $n - 1$:

If array $[j] < \text{array} [\text{min index}]$

min index = j .

If min index $\neq i$

swap array $[i]$ and array $[\text{min index}]$

conclusion: The implemented different searching and sorting method in the student database.

FAQs.

Q.1 What is the meaning of database? How to maintain in?

A database is an organised collection of structure information or data stored electronically allowing efficiency retrieval insertion, modification and delete.

2) What are the applications of searching?

Ans Searching algorithms are essential in computer science for locating target elements within a dataset. Key applications include:

- Data base queries: Retrieving specific user records, bank transactions or inventory items by primary key or ID.

- Search engine: Finding relevant web pages, records, bank transactions or inventory items by primary key or ID.

3) Compare linear search and Binary search.

Linear search

Binary Search

1) works on sorted or unsorted data

1) Requires data to be strictly sorted

2) Checks elements sequentially from start to end

2) Use divide and conquer to repeatedly halve the search space

3) $O(1)$ [OCTC]

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Q 4) What are applications of sorting

- Sorting reorganize data into a specific order optimizing other computation task.
- Efficient searching: Enables logarithmic - time operation like binary search large data used.
- Data visualization & Analysis: Ranking, sales ranking; Students merit list.
- E-commerce filtering: Sorting problem by price rating or popularity.

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